

Fields in FFGFT

Energy, Flux and Geometry

*Poynting vector, Goubau line and
time-mass duality*

Johann Pascher

ORCID: 0009-0000-6518-4064

<https://github.com/jpascher/T0-Time-Mass-Duality>

19 September 2026

Doc. 366 — FFGFT Corpus

Keywords: electromagnetism · antenna engineering · Goubau line · Poynting vector · field theory · time–mass duality · FFGFT · mode compatibility · crystal lattice · orbital resonances

DOI: Zenodo FFGFT corpus v1.3.x

Licence: CC BY 4.0 — Johann Pascher, 2026

Contents

Fields in FFGFT: Energy, Flux and Geometry 1

 What this document explains — and what it does not 2

 List of symbols 3

 1 Starting point: fields or current? 5

 2 The Poynting vector as the primary description 5

 3 The Goubau line: fields without a return conductor 6

 4 FFGFT units: current is mass, voltage is energy 7

 5 Field lines as geometric objects on $T^4/\mathbb{Z}_3$ 7

 6 Antenna engineering as mature field theory 9

 7 Multiple field carriers in space: boundary conditions, coupling and apparent
 superposition 16

 8 Particles as field nodes: near and far field at localised field structures . . . 19

 9 Scale-invariance of the coupling condition: atoms, elements and planetary
 orbits 23

 10 Duality: field and particle, energy and current 27

 11 Exact agreement: Poynting integral and power formula 28

 12 Summary 28

References 29

Fields in FFGFT: Energy, Flux and Geometry

Abstract

This document develops a unified approach to the field structure of FFGFT, starting from an unexpected experimental result and ending at the orbital resonances of the planets. The entry point is deliberately concrete: the Goubau line experiment — energy transport along a single wire without a return conductor — forces the question of where energy actually flows to be taken seriously. The answer has been fully present in classical field theory since Poynting (1884); it is regularly obscured in introductory courses, however, by the useful but simplified picture of current and return path.

The goal is not to claim new physics. The goal is to make visible that antenna engineering, atomic physics, crystallography and celestial mechanics all realise structurally the same principle: the geometry of the carrier determines the permitted coupling angles. Coupling between two field systems is only possible where their mode structures are compatible. Incompatible modes do not couple — regardless of the field strength.

In FFGFT this connection is not an analogy but a consequence of the fundamental equation $\tilde{T} \cdot m = 1$ on the compact carrier $T^4/\mathbb{Z}_3$. The winding numbers (n_θ, n_ϕ) on this carrier are the topological invariants behind the selection rules of atomic physics, the Bragg conditions of crystallography and the integer orbital resonances of celestial mechanics. In FFGFT units ($c = \hbar = \alpha = 1$): $I = m$, $V = E$, $P = E \cdot m$ — exactly, not approximately.

Antenna engineering serves as the guide because it is the most mature practical application of electromagnetic field theory: the coupling conditions are not abstract there but codified in standards, measurement protocols and commercial simulators. What is known there as the Friis equation, impedance matching and array factor appears in atomic physics as a selection rule, in the crystal as the Bragg condition and in the solar system as the Laplace resonance — the same structure, different carriers.

What this document explains — and what it does not

Four questions are addressed, developed in this order:

1. Where does energy flow? Not in the conductor, but in the field around it. The Goubau experiment forces this answer experimentally; the Poynting vector delivers it mathematically exactly. Current and voltage are projections, not primary quantities (§ 1–3).

2. What does an antenna do to the field? It actively reshapes it — in transmission by imposing boundary conditions, in reception by acting as a selective mode filter. Passive in a field it scatters; in near-field proximity two antennas mutually deform each other. This is not superposition but mutual deformation of mode structures (§ 4–9).

3. What is a particle in this picture? A stable field node — a localised mode concentration on the carrier. The near field of the particle is the Coulomb field; the far field arises only under acceleration or a state transition. The selection rules of atomic physics are the mode-compatibility conditions of this node (§ 10).

4. Does the principle hold on all scales? Yes — from the antenna through the atom and the crystal to the planetary orbits. The coupling condition is always the same: integer phase compatibility of the mode spectra. The scale changes; the structure of the condition does not (§ 11).

What this document does *not* explain: why the carrier is $T^4/\mathbb{Z}_3$ (that is **[POSIT]** in Doc. 365); why $\xi = 4/30\,000$ (that follows from Galois group orders, Doc. 338/365); how concrete masses follow from mode compatibility (Doc. 314, 317–323). Those documents are the sources cited here. This document is the approach from the concrete to the structural.

Reading note: Readers familiar with antenna engineering may skim § 1–3 and enter at § 4. Those who know the FFGFT background will find the connection to mode structure in § 5, § 10 and § 11.

List of symbols

	[B]	algebraically proven
	[K]	numerically confirmed
	[S]	structurally plausible
Epistemic markers:	[POSIT]	motivated input
	[E]	established mathematic- s/physics
	[X]	checked and negative

Symbol	Meaning
<i>Fields and energy</i>	
$F^{\mu\nu}$	electromagnetic field strength tensor
$\mathbf{E}, \mathbf{B}$	electric and magnetic field
$\mathbf{H}$	magnetic field strength ($\mathbf{H} = \mathbf{B}/\mu_0$ in vacuum)
$\mathbf{S} = \mathbf{E} \times \mathbf{H}$	Poynting vector (energy flux density, W/m^2)
j^ν	electromagnetic four-current
u	electromagnetic energy density
ϵ_0, μ_0	permittivity of free space, permeability of free space
λ	wavelength ($\lambda = c/f$)
$\omega = 2\pi f$	angular frequency
$k = 2\pi/\lambda$	wave number
θ	elevation angle (radiation pattern) or glancing angle (Bragg)
ψ	polarisation mismatch angle between transmit and receive antenna
<i>FFGFT quantities</i>	
$\tilde{T}$	intrinsic time scale; $\tilde{T} \cdot m = 1$
$\xi = 4/30\,000$	dimensionless scaling parameter of FFGFT
$T^4/\mathbb{Z}_3$	compact orbifold carrier of FFGFT
(n_θ, n_φ)	winding numbers on T^4 ; topological invariants
<i>Antenna engineering</i>	
R_{rad}	radiation resistance (power radiated into space)
G	antenna gain (relative to isotropic radiator)
D	directivity (gain without losses)
A_{eff}	effective aperture; $A_{\text{eff}} = (\lambda^2/4\pi)G$
η_A	aperture efficiency ($0 < \eta_A \leq 1$)
G_T, G_R	gain of transmit and receive antenna (Friis equation)
$\Delta\theta_{3\text{dB}}$	half-power beamwidth
Z_{ant}	input impedance of the antenna
X_{ant}	reactance (imaginary part of Z_{ant})

ϕ_n	feed phase of the n -th element in a phased array
$AF(\theta)$	array factor (interference pattern of the phased array)
ϵ_r	relative permittivity (dielectric constant)
ϵ_{eff}	effective permittivity of a microstrip line
ΔL	fringe correction of the patch antenna length
k_{SPP}	wave number of a surface plasmon polariton
ϵ_m, ϵ_d	(complex) permittivity of metal and dielectric (SPP)

Atomic physics and scales

$\hbar$	reduced Planck constant
$\alpha \approx 1/137$	fine-structure constant; in FFGFT units $\alpha = 1$
n, l, m	principal, orbital angular momentum, and magnetic quantum numbers
E_n	energy levels of the hydrogen atom; $E_n = -13.6 \text{ eV}/n^2$
$r_e \approx 2.82 \text{ fm}$	classical electron radius
$\lambda_C = \hbar/m_e c$	Compton wavelength of the electron ($\approx 386 \text{ fm}$)
$a_0 \approx 52\,917 \text{ fm}$	Bohr radius
σ_T	Thomson cross-section; $\sigma_T = \frac{8\pi}{3} r_e^2$
S_{AB}	overlap integral of two atomic orbitals ψ_A, ψ_B
Q	normal coordinate of a molecular vibration
$\vec{\mu}$	electric dipole moment of a molecule
d	lattice plane spacing in a crystal (Bragg condition)
n_1, n_2, n_3	mean motions (orbital frequencies) of the Galilean moons

1 Starting point: fields or current?

Every introductory course on electricity emphasises the principle: current needs a return path. Electrons leaving one conductor must return on the other. This picture is entirely correct for technical purposes — it describes circuits precisely. It conceals a deeper question, however: where does the energy actually flow?

The Goubau experiment **[Q]** (G. Goubau, 1950) demonstrates the answer experimentally. A high-frequency signal source (~ 900 MHz) feeds a single isolated wire via a conical transition (launcher). At the other end an identical cone (catcher) converts the signal back into a coaxial line. Between the two cones there is no second conductor, no return path in the classical sense — yet the arrangement transfers the signal with low losses **[1]**.

A physical interruption of the field (for instance a hand placed over the wire) immediately interrupts the transfer. The energy flows in the field around the wire, not in the wire itself.

2 The Poynting vector as the primary description

The energy flux density of electromagnetic fields is the Poynting vector **[3, 4]**:

$$\mathbf{S} = \mathbf{E} \times \mathbf{H}.$$

Its direction (right-hand rule: $\mathbf{E} \rightarrow \mathbf{H}$, thumb in the direction of flux) indicates where energy is transported. The flux density integrated over the cross-section of a cable gives exactly the electrical power **[E]**:

$$P = \oint_A \mathbf{S} \cdot d\mathbf{A} = V \cdot I.$$

This is not an approximation, not a coincidence — it is an exact identity **[4]**. Current and voltage are macroscopic projections of the field process, not the primary carriers of energy.

Consequence. For a coaxial line the $\mathbf{E}$ field lines run radially from the inner conductor to the outer shield; the $\mathbf{H}$ field lines form concentric rings. The cross product $\mathbf{E} \times \mathbf{H}$ points axially — along the line. Energy flows in the dielectric between inner and outer conductor, not in the copper.

3 The Goubau line: fields without a return conductor

The Goubau line (G-line) has no outer conductor. The field configuration differs from the coaxial line only gradually when measured close enough to the wire [E].

Field configuration

A voltage pulse propagating on the wire creates alternating positive and negative potential regions along the conductor. The electric field lines run between these regions — not from wire to return conductor, but from one section of the wire to the next. The dielectric (the red PVC insulation) slows the propagation of the fields and bends the field lines towards the wire axis, just as glass bends light rays [1, 2]. This mechanism keeps the fields close to the wire — it is not cosmetic but physically necessary for practical operation.

The cone transition as impedance matching

The cones (launcher / catcher) match the wave impedance of the coaxial line ($50\ \Omega$) to the characteristic impedance of the G-line (typically a few $100\ \Omega$) [E]. Inside the cone the field lines expand from the coaxial pattern to the G-line pattern. At the opening of the cone the field lines snap and attach to the wire. The return path for the current in the coaxial cable is provided by the displacement currents inside the cone — the complex fields in the cone supply exactly the right forces to drive electrons in the outer surface.

Losses

Because current flows in the wire, I^2R losses occur — exactly as in a conventional two-wire line. This shows: the G-line is not a pure waveguide problem. The energy flux has a field component (Poynting) and a conduction component (Ohm); both describe the same physics from different perspectives.

4 FFGFT units: current is mass, voltage is energy

In FFGFT units ($c = \hbar = \alpha = 1$, Doc. 011; Doc. 365 Layer 2) electromagnetic quantities simplify to geometric expressions in mass and time **[K]**:

$$e = 1 \quad (\text{elementary charge dimensionless}),$$

$$V = E/e = E \quad (\text{voltage} = \text{energy}),$$

$$I = e/T = 1/T = m \quad (\text{current} = \text{mass, via } \tilde{T} \cdot m = 1).$$

The power equation then reads:

$$P = V \cdot I = E \cdot m,$$

a relation that follows directly from the fundamental equation $\tilde{T} \cdot m = 1$ (Doc. 365 Layer 1) **[K]**.

Maxwell equations in FFGFT units. In FFGFT units (Doc. 365 § 2) the Maxwell equations take the form:

$$\partial_\mu F^{\mu\nu} = j^\nu, \quad \partial_{[\mu} F_{\nu\rho]} = 0.$$

No prefactors, no coupling constants other than j^ν as a geometric object. The coupling is absorbed in the unit choice, not eliminated; the physical content is identical to the SI system.

5 Field lines as geometric objects on $T^4/\mathbb{Z}_3$

In FFGFT the physical carrier is the compact orbifold $T^4/\mathbb{Z}_3$ (Doc. 365 Layer 1) **[POSIT]**. Fields are sections of vector bundles over this carrier; their periodicity conditions enforce discrete mode spectra.

Field lines and winding numbers

On a torus T^4 non-trivial field configurations have topological invariants — winding numbers (n_θ, n_φ) that record how many times a field line traverses the torus in each direction (Doc. 314; Doc. 247) **[B]**. These winding numbers classify the modes and determine — via $\tilde{T} \cdot m = 1$ — the mass scale of the respective sector. A field line in this description is not an auxiliary construct but a topological object with physical relevance.

Field lines in the G-line context

The behaviour of field lines in the Goubau line — expansion in the cone, snapping at the opening, attachment to the wire — is from this perspective a macroscopic example of the energy-flux minimisation principle. At every position and time the flux chooses the path that keeps the Poynting vector consistent with the carrier. This is not mysterious behaviour but the continuum version of topological optimisation on $T^4/\mathbb{Z}_3$.

Field shape and carrier: no one-to-one correspondence

The fundamental equation $\tilde{T} \cdot m = 1$ says nothing about the *shape* of a field in space. It specifies which mass scales belong to which time scales — not how the field looks geometrically. The compact carrier $T^4/\mathbb{Z}_3$ is the topology that prescribes the permitted boundary conditions; the actual field configurations on and around this carrier are different from it and variously deformed depending on the situation.

A comparison clarifies this: the length of a guitar string determines the permitted frequency spectrum — but the sound wave in the surrounding space is not string-shaped. Equally, $T^4/\mathbb{Z}_3$ enforces the mode spectrum; what appears as a field in the three-dimensional observation space is a projection of it, depending on the concrete boundary conditions, the ambient geometry and the mode occupation. Close to the conductor the G-line field is cylindrical, further away elliptically deformed, at inhomogeneities asymmetric — none of this is torus-shaped. The torus is the internal geometric carrier, not the shape of the field.

Interaction: carrier determines permitted coupling angles

When two fields from different carriers overlap in space, their interaction is not simple superposition. Each field carries its own mode structure, fixed by the winding numbers (n_θ, n_φ) of its carrier. Coupling is only possible where the phase conditions of both mode spectra are compatible — where the winding numbers fit together as integers. The geometry of the carrier thus determines the *angles* of interaction: under which phase relations energy transfer occurs and under which it does not [S].

This is worked out quantitatively in the corpus: the Galois lattice condition from Doc. 328 (coupling regimes) and Doc. 343 § G' specifies that coupling is only effective at frequency ratios in the Galois lattice $\{2, 3, 5, 7, 11, 13\}$; the resolution floor 100ξ is the sharpest bound below which two modes are no longer distinguishable as separate. The apparently free shape of a field in space is the result of these compatibility conditions, projected onto the observation space **[S]**.

6 Antenna engineering as mature field theory

Antenna engineering is the most mature practical application of electromagnetic field theory. Decades of measurements, standards and calculation procedures are available; the coupling conditions are not abstract but anchored in handbooks, measurement protocols and commercial simulators. It is precisely for this reason the most suitable introduction to the general principle: the geometry of the carrier determines the permitted coupling angles.

Fundamental quantities and their field interpretation

Every antenna is a transducer between a guided wave (coaxial cable, waveguide) and a free wave in space. The decisive quantities are **[E]**:

- **Radiation resistance** R_{rad} : The resistance the antenna presents to the guided wave because it radiates energy into space. For the half-wave dipole: $R_{\text{rad}} \approx 73.1 \Omega$. In FFGFT units ($I = m$, $V = E$): $R_{\text{rad}} = P/I^2 = E/m$. The radiation resistance is thus an energy-per-mass relation — a direct echo of $\dot{T} \cdot m = 1$ **[S]**.
- **Gain** G and **directivity** D : Gain gives the factor by which the radiation intensity in the main direction exceeds that of an isotropic radiator. It is a pure field-geometry problem: how do the phase conditions of the carrier (antenna length, element spacing, feed phase) concentrate the Poynting vector in a preferred direction? For the half-wave dipole: $D = 1.64$ (2.15 dB).
- **Effective aperture** A_{eff} : The area from which a receiving antenna draws energy. Relation to gain **[E]**:

$$A_{\text{eff}} = \frac{\lambda^2}{4\pi} G.$$

This formula is pure field geometry: wavelength and gain determine the effective area, not the physical size of the antenna.

Friis transmission equation: coupling of two carriers

The Friis equation describes energy transfer between two antennas in free space [7, 6]:

$$\frac{P_R}{P_T} = G_T G_R \left(\frac{\lambda}{4\pi r} \right)^2 \cos^2 \psi,$$

where P_T is the transmitted, P_R the received power, G_T, G_R the gains of transmit and receive antennas, r the distance and ψ the polarisation mismatch angle.

Every factor of this equation corresponds to a compatibility condition between the carriers:

Factor	Meaning	FFGFT
G_T	transmit cussing	fo- source mode geom.
G_R	receive cussing	fo- receive mode geom.
$(\lambda/4\pi r)^2$	free-space loss	propagation factor
$\cos^2 \psi$	polarisation overlap	winding com- pat.

The $\cos^2 \psi$ factor is the most direct expression of the principle: at orthogonal polarisation ($\psi = 90^\circ$) transfer is zero — not because of absorption or loss but because the mode geometries are incompatible. No transmitter of arbitrary power overcomes this. This is identical to the statement about winding numbers: if they do not match, no energy transfer occurs [S].

Impedance matching as a phase condition

For an antenna to radiate at maximum efficiency its input impedance must be the complex conjugate of the source impedance (conjugate matching) [E]:

$$Z_{\text{ant}} = R_{\text{rad}} + jX_{\text{ant}}, \quad Z_{\text{src}} = R_{\text{src}} - jX_{\text{ant}}.$$

The imaginary part — the reactance — measures stored field energy near the antenna (near field, reactive power). If it is not compensated, energy oscillates between source and antenna near field without being radiated. This is the same phenomenon as the reactive power on the G-line (§ 3): energy pulses in the near field without net transport.

For the G-line the cone transition was the matching structure (50 Ω to $\sim 300 \Omega$); for antennas this is handled by $\lambda/4$ transformers, baluns or matching networks. The mechanism is structurally identical: transition between two mode spectra through geometrically enforced phase condition.

Antenna types as mode geometries

Different antenna types realise different mode geometries. Each type is an independent carrier with a characteristic mode spectrum; the radiation pattern in the far field is the projection of this spectrum onto the observation space.

1. Short dipole ($\ell \ll \lambda$). The short dipole is the simplest limiting case: a straight conductor whose length is far below the wavelength. The current distribution is approximately triangular; the far field is analytically exact [E]:

$$E_\theta \propto \frac{I_0 \ell}{\lambda r} \sin \theta, \quad F(\theta) = \sin^2 \theta.$$

The radiation pattern $F(\theta) = \sin^2 \theta$ is a torus shape around the dipole axis: maximum radiation perpendicular to the axis ($\theta = 90^\circ$), zero along the axis ($\theta = 0^\circ, 180^\circ$). Directivity $D = 1.5$ (1.76 dBi). Radiation resistance:

$$R_{\text{rad}} = 80\pi^2 \left(\frac{\ell}{\lambda}\right)^2 \approx 790 \Omega \cdot \left(\frac{\ell}{\lambda}\right)^2.$$

For $\ell = 0.1\lambda$: $R_{\text{rad}} \approx 7.9 \Omega$ — very low, hence severe mismatch to 50 Ω systems. The mode structure is simple: a single electric dipole mode whose winding geometry enforces the torus shape of the radiation pattern.

2. Half-wave dipole ($\ell = \lambda/2$). The half-wave dipole is the most important reference radiator in antenna engineering. The current

distribution is sinusoidal; the radiation pattern is slightly narrower than the short dipole **[E]**:

$$F(\theta) = \frac{\cos^2\left(\frac{\pi}{2} \cos \theta\right)}{\sin^2 \theta}.$$

Directivity $D = 1.64$ (2.15 dBi). Radiation resistance $R_{\text{rad}} = 73.1 \Omega$ — nearly ideal for 50Ω or 75Ω coaxial cable (mismatch $< 2.5 : 1$ without matching network). The half-wave resonance condition ($\ell = \lambda/2$) enforces exactly the phase relationship at which radiation resistance and feed resistance enter a real ratio: the imaginary part of the input impedance vanishes at resonance. This is an integer mode-compatibility condition — the length must be a half-integer multiple of the wavelength.

3. Monopole ($\ell = \lambda/4$) over ground plane. A quarter-wave monopole over an infinitely extended conducting plane is equivalent to the half-wave dipole by image theory: the ground plane supplies the mirror image. The radiation pattern is the lower hemisphere of the dipole torus, reflected upward; all radiated power concentrates in the upper half-space **[E]**:

$$D_{\text{mono}} = 2 D_{\text{dipole}} = 3.28 \quad (5.15 \text{ dBi}).$$

Radiation resistance $R_{\text{rad}} = 36.6 \Omega$ (half that of the dipole, since only half the space). Practical implementations: car antenna, mobile mast **[6]**. The gain increase over the dipole arises not from more input power but from the geometric mirroring condition — doubling the solid angle contributing to radiation is a pure mode-geometry question.

4. Yagi–Uda antenna. The Yagi–Uda antenna consists of a driven dipole (radiator), a slightly longer passive reflector and a series of shorter passive directors **[8, 6]**. All parasitic elements are excited by induction; their length determines the phase of the induced current:

- Reflector ($\ell \approx 0.5\lambda + 5\%$): capacitive, re-radiates with phase lag $\Rightarrow$ cancellation to the rear.
- Directors ($\ell \approx 0.5\lambda - 5\%$): inductive, re-radiate with phase lead $\Rightarrow$ constructive interference forward.

Gain as a function of the number of directors n (rule of thumb) **[E]**:

$$G \approx 10 \log_{10}(0.8n + 1) \text{ dBd} \quad (n = \text{number of directors}).$$

Example values: 3 directors $\Rightarrow G \approx 7 \text{ dBd}$, 10 directors $\Rightarrow G \approx 10 \text{ dBd}$.
The half-power beamwidth (in the E plane) is approximately:

$$\Delta\theta_3 \text{ dB} \approx \frac{52^\circ}{\sqrt{G_{\text{lin}}}},$$

where G_{lin} is the linear gain factor. The radiation pattern narrows with increasing element count — the carrier becomes more selective in its coupling angles.

5. Horn antenna. The horn antenna is the transition from waveguide to free space. The waveguide supports a discrete mode structure (TE_{10} fundamental mode for rectangular waveguide); the horn adiabatically expands this mode into a plane wave in the far field [6, 9]. The half-power beamwidth depends directly on the aperture dimensions:

$$\Delta\theta_E \approx 56^\circ \frac{\lambda}{a}, \quad \Delta\theta_H \approx 67^\circ \frac{\lambda}{b},$$

where a is the E -plane aperture and b the H -plane aperture. Gain of a pyramidal horn with aperture $A = a \times b$:

$$G = \eta_A \frac{4\pi A}{\lambda^2}, \quad \eta_A \approx 0.5\text{--}0.7.$$

The aperture efficiency $\eta_A < 1$ arises from the non-uniform phase distribution in the aperture (spherical rather than plane wave at the horn mouth). This is directly the mode incompatibility between waveguide near field and far-field plane wave: the better the phase matching (flatter aperture), the higher η_A .

6. Parabolic antenna. A parabolic antenna of diameter D and focal length f focuses incident microwaves at the focus. In transmission a primary radiator at the focus converts a spherical wave into a plane wavefront at the aperture plane. The far field is the Fourier transform of this aperture distribution **[E]**. For a circular, uniformly illuminated aperture:

$$F(\theta) = \left[\frac{2 J_1(\pi D \sin \theta / \lambda)}{\pi D \sin \theta / \lambda} \right]^2,$$

where J_1 is the first-order Bessel function [6]. Gain and half-power beamwidth:

$$G = \eta_A \frac{\pi^2 D^2}{\lambda^2}, \quad \Delta\theta_{3\text{ dB}} \approx 70^\circ \frac{\lambda}{D}.$$

For a 3-m parabolic antenna at 10 GHz ($\lambda = 3\text{ cm}$): $G \approx 10^4$ (40 dBi), $\Delta\theta \approx 0.7^\circ$. The Bessel function in the radiation pattern is not an approximation but the exact consequence of the circular symmetry of the aperture: the mode geometry (circle) completely determines the coupling pattern in the far field.

7. Patch antenna (microstrip). A rectangular patch antenna of length L and width W over a ground plane (dielectric ϵ_r , thickness h) resonates when L equals half an effective wavelength [9, 6]:

$$L = \frac{\lambda}{2\sqrt{\epsilon_{\text{eff}}}} - 2\Delta L,$$

where ϵ_{eff} is the effective permittivity and ΔL the fringe correction:

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + \frac{12h}{W}\right)^{-1/2}.$$

The radiation pattern is broadside in the H plane (no focussing), narrow in the E plane; typical gain 5–8 dBi. The permitted modes are completely determined by geometry and material (L, W, h, ϵ_r) — no free parameters. Coupling to an external wave is only possible when its polarisation and angle of incidence are compatible with the resonance mode of the patch. Cross-polarisation isolation ($> 30\text{ dB}$) shows: orthogonal polarisation practically does not couple, even when the frequency matches exactly.

8. Phased array. A phased array consists of N identical radiators with individually controlled feed phase ϕ_n . The array pattern is the product of the element radiation pattern $F_{\text{elem}}(\theta)$ and the array factor [10, 6]:

$$AF(\theta) = \sum_{n=0}^{N-1} \exp[j(nkd \cos \theta + \phi_n)],$$

where $k = 2\pi/\lambda$ is the wave number and d the element spacing. By choosing $\phi_n = -nkd \cos \theta_0$ the main beam steers to angle θ_0 —

electronic beam steering without mechanical movement. Gain for N elements with spacing $d = \lambda/2$:

$$G_{\text{array}} = N \cdot G_{\text{elem}}.$$

The phased array is the purest example of the coupling-angle principle: the feed phases ϕ_n are integer multiples of a basic phase step — they determine under which angle constructive interference occurs. Every other angle combination gives destructive interference. In FFGFT language: the ϕ_n are discrete winding phase values; the array factor selects the compatible coupling angles **[S]**.

Summary table.

Type	Gain	$\Delta\theta$	Mode geom.
Short dipole	1.76 dBi	90°	electric dipole mode
Half-wave dipole	2.15 dBi	78°	resonance $\ell = \lambda/2$
Monopole ($\lambda/4$)	5.15 dBi	46°	image mode over ground
Yagi (5 dir.)	≈ 12 dBd	$\approx 30^\circ$	interference array
Horn	15–25	3–15°	Aperture
Parab. 3m/10GHz	40 dBi	0.7°	Fourier
Patch	5–8 dBi	60°–120°	resonance mode in dielectric
Phased (N)	NG_e	$\propto 1/\sqrt{N}$	steered

In every case: the physical geometry of the antenna determines the permitted mode spectrum; what appears in the far field as a radiation pattern is the projection of this spectrum onto the observation space. No antenna type transfers energy at angles that contradict its mode geometry — not approximately but structurally. This is the technically mature, quantitatively anchored form of the principle formulated in FFGFT as winding-number compatibility **[S]**.

Reciprocity: transmitter and receiver as dual descriptions

The reciprocity principle of antenna engineering states: gain, radiation pattern and impedance of an antenna are identical in transmit and receive mode [6, 4]. This symmetry is not coincidental — it reflects the time-reversal symmetry of the Maxwell equations, and with it the duality of field source and field sink. In FFGFT language: transmitter and receiver are reciprocal projections of the same field process, just as $\hat{T}$ and m are reciprocal representations of the same object [S].

7 Multiple field carriers in space: boundary conditions, coupling and apparent superposition

The superposition principle of classical field theory states: in a linear medium the fields of independent sources add linearly. Each partial field propagates as if the other were not present. This is an excellent approximation for widely separated, weakly coupling sources — and it works very well in practice. It conceals, however, what physically actually occurs.

The total field as the consistent solution of all boundary conditions

Every object in a field — an antenna, a conductor, a dielectric, an atom — is itself a field carrier with its own mode structure. It imposes boundary conditions: the tangential components of the electric field must vanish at every conducting surface; the normal components of flux density must be continuous across interfaces [E]. The total field is not the sum of the partial fields — it is the unique consistent field satisfying all boundary conditions simultaneously.

Superposition is the approximation that arises when the boundary conditions of individual objects are spatially far enough apart not to influence each other. As soon as objects enter near-field range, this approximation breaks down — not gradually but structurally.

Near field: where boundary conditions collide

The near-field zone of an antenna extends to roughly $r < 2D^2/\lambda$ (Fraunhofer boundary), where D is the antenna size [E]. In this zone

the field cannot be described as a plane wave; it contains reactive components (stored field energy oscillating back and forth without being radiated). Bringing a second antenna into this zone causes the following:

- The input impedance of both antennas changes — the resonance frequency shifts, the radiation resistance changes. This is not field addition but mutual deformation of the mode structures.
- The radiation pattern of each antenna changes because the boundary condition of the other deforms the field distribution in the shared near space.
- Energy can be transferred directly between near fields — without radiation into the far field. This is the principle of inductive coupling (transformer, Qi charging), capacitive coupling and resonant near-field coupling (Kurs et al. 2007 [12], wireless transfer over metres).

Mutual coupling in antenna arrays

In a phased array with element spacing $d < \lambda$ the mutual coupling is not negligible [10, 9]. The input impedance of the n -th element depends on all others:

$$Z_{nn}^{\text{active}} = Z_{nn}^{\text{isolated}} + \sum_{m \neq n} Z_{nm} \frac{I_m}{I_n},$$

where Z_{nm} is the mutual impedance between elements n and m . The overall behaviour of the array is only describable by solving the complete system of equations — not by addition of single-element fields. In practice mutual coupling is compensated by optimising element geometry and feed phases; it is a design problem that industry solves with full electromagnetic simulators (MoM, FEM, FDTD).

Cavity resonator: the total field as an eigenmode

A metal cavity allows only discrete resonance frequencies determined by its geometry [9, 4]. For a rectangular cavity of dimensions $a \times b \times c$:

$$f_{mnp} = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2 + \left(\frac{p}{c}\right)^2}.$$

Placing an object inside the cavity — a dielectric block, a conductor, biological tissue — shifts all resonance frequencies. The entire mode spectrum of the cavity changes, not only the modes that directly

touch the object. The total field is the eigenmode of the composite system (cavity + object), not the sum of the empty-cavity eigenmode and a scattered field. This shift is measurable and exploited in microwave dielectric spectroscopy (cavity perturbation method).

Faraday cage: boundary conditions extinguish a field

The Faraday cage is the most extreme case: a closed conducting shell enforces vanishing tangential electric field on its inner surface. The external field continues unchanged; the internal field is forced to zero **[E]**. The total field is the unique solution of the Maxwell equations with this boundary condition — not superposition but cancellation through enforced compatibility.

Incomplete shielding (gaps, finite conductivity) shows the principle in attenuated form: fields penetrate but only with mode geometries permitted by the gap geometry. A slot of width w allows fields with $\lambda > 2w$ to pass through strongly attenuated (cut-off condition of the slot waveguide). This is again mode compatibility: only fields whose wavelength is compatible with the slot geometry couple through.

Plasmons: field and matter inseparable

At the interface between metal and dielectric, surface plasmon polaritons (SPP) arise — coupled modes of electromagnetic field and collective electron oscillation [11]. The dispersion relation is:

$$k_{\text{SPP}} = \frac{\omega}{c} \sqrt{\frac{\epsilon_m \epsilon_d}{\epsilon_m + \epsilon_d}},$$

where ϵ_m is the (complex) permittivity of the metal and ϵ_d that of the dielectric. The SPP is neither pure electromagnetic field nor pure charge carrier oscillation — it is a joint mode of both carriers. Superposition would be meaningless: the field exists only in coupling with matter.

This is the sharpest available evidence that “multiple fields in space” are not independent objects. The mode structure of the total system is the physical reality; decomposition into partial fields is a computational method that works under certain conditions.

FFGFT interpretation: one field, many projections

In FFGFT there is only one field on the carrier $T^4/\mathbb{Z}_3$ **[POSIT]**. What appears as “multiple fields” are projections of different mode groups onto the observation space. The apparent superposition is a representational question.

When two objects with different mode structures interact, the result is not the sum of their fields but the joint eigenmode of the composite carrier — determined by the compatibility of the winding numbers of both carriers. This is the same principle that appears in antenna engineering as mutual coupling, in cavity physics as cavity perturbation and in nanophotonics as plasmon coupling **[S]**.

The complexity of field interaction is therefore not a complication that sufficient computational effort can remove — it is the direct consequence of every carrier imposing boundary conditions on the total field. The more carriers in space, the more simultaneously required compatibility conditions, the more complex the total field. This is not superposition — this is geometry.

8 Particles as field nodes: near and far field at localised field structures

In classical physics a particle is an object that *has* a field — the charge generates the Coulomb field, the mass generates the gravitational field. In FFGFT this description is inverted: a particle is a stable, localised field configuration — a node in the global field on the carrier $T^4/\mathbb{Z}_3$ **[POSIT]**. Mass is not an intrinsic property of the node but follows from the fundamental equation $\tilde{T} \cdot m = 1$: the time scale of the field configuration determines the mass (Doc. 365 Layer 1, Doc. 124) **[K]**.

This inversion has direct consequences for the structure of the field around a particle — and it makes the connection to antenna engineering visible.

Stationary particle: pure near field

A stationary point charge e generates the Coulomb field **[E]**:

$$\mathbf{E}(r) = \frac{e}{4\pi\epsilon_0} \frac{\hat{r}}{r^2}, \quad \mathbf{B} = 0.$$

In FFGFT units ($e = 1$, $4\pi\epsilon_0 = 1$): $V(r) = -1/r$ [21]. This field falls off as $1/r^2$ (field strength) or $1/r$ (potential), carries no energy into the far field and is fully reactive — identical to the near field of a stationary antenna. The Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ is zero for a stationary charge: no energy flux, only stored field energy in the node structure.

The Coulomb field is the near field of the particle. It deforms the space around the node, imposing boundary conditions on all other fields in its vicinity — exactly as a passive antenna deforms the field in its surroundings without transmitting.

Accelerated particle: near field and radiated far field

As soon as the charge is accelerated, a far field arises — a detaching electromagnetic wave. The Larmor formula gives the radiated power [5, 4]:

$$P = \frac{e^2 a^2}{6\pi\epsilon_0 c^3} = \frac{2}{3} \frac{e^2 a^2}{c^3} \quad (\text{in Gaussian units}),$$

where a is the acceleration. In FFGFT units ($c = e = 1$): $P = \frac{2}{3} a^2$.

The far field falls off as $1/r$ (not $1/r^2$), permanently transports energy away and is propagating — identical to the far field of a driven antenna. The boundary between near and far field lies at $r \sim \lambda/2\pi$ (transition zone), where $\lambda = c/f$ and f is the characteristic frequency of the acceleration.

This is the antenna principle at the particle level:

- Stationary charge = undriven antenna: near field only, no radiation.
- Uniformly moving charge = antenna on direct current: no alternating current, no radiation (Cherenkov radiation is a special case for $v > c/n$).
- Accelerated charge = driven antenna: near field plus radiated far field, power $\propto a^2$.

Atomic transition: quantised far field

In the atom the permitted node structures are discrete through the quantum numbers (n, l, m) — these are the permitted modes of the Coulomb potential, i.e. the eigenmodes of the atomic carrier. A transition between two states is a change in node structure; a photon is emitted — the radiated far field carries away a discrete energy $\Delta E = \hbar\omega$. The selection rules [14, 13]:

$$\Delta l = \pm 1, \quad \Delta m = 0, \pm 1$$

are compatibility conditions: only transitions in which the change of mode geometry is compatible with the geometry of a dipole photon are permitted. Quadrupole and higher transitions ($\Delta l = \pm 2$) are possible but suppressed by a factor $\alpha^2 \approx 10^{-4}$ — because the mode compatibility is poorer.

In FFGFT language: the winding numbers of the initial and final state must differ by exactly the winding numbers of a photon. The selection rules are not postulated but follow from the geometry of the carrier **[S]**.

The “size” of a particle: extent of the node

A particle as field node has no sharp geometric boundary. The “size” is the spatial extent of the node structure, i.e. the region where the near field dominates over the background field. For the electron these are the classical electron radius **[E]**:

$$r_e = \frac{e^2}{4\pi\epsilon_0 m_e c^2} \approx 2.82 \text{ fm},$$

the Compton wavelength $\lambda_C = \hbar/m_e c \approx 386 \text{ fm}$ and the Bohr radius $a_0 \approx 52\,917 \text{ fm}$. These three lengths are not arbitrary but linked through α **[5, 21]**:

$$r_e = \alpha \lambda_C / 2\pi = \alpha^2 a_0.$$

In FFGFT units ($\alpha = 1$) these three lengths collapse to one — the hierarchy arises only through the SI embedding via ξ (Doc. 365 Layer 3) **[K]**.

The node is therefore not point-like — it has a structure extending over several length scales. What appears as a “point” is the projection of this structure onto a measurement coarse enough not to resolve the inner structure.

Comparison table: antenna and particle

Property	Antenna	Particle
Near field	reactive, $E \propto 1/r^2 - 1/r^3$	Coulomb field $E \propto 1/r^2$
Far field	radiated wave, $E \propto 1/r$	photon under acceleration
Radiation	only on current change	only under acceleration/transition
Mode structure	antenna geometry	quantum numbers (n, l, m)
Selection rule	impedance compatibility	$\Delta l = \pm 1, \Delta m = 0, \pm 1$
"Size"	physical dimension	node extent r_e, λ_C, a_0
Stationary	no far field	no far field (Coulomb only)
Passive in field	scattering, shadowing	polarisability, cross-section

Cross-section as a measure of node coupling

A particle in an external field scatters the incoming wave — it deforms the field in its vicinity, exactly like a passive conductor. The Thomson cross-section of the electron [5]:

$$\sigma_T = \frac{8\pi}{3} r_e^2 \approx 6.65 \times 10^{-29} \text{ m}^2$$

is the effective area from which the electron draws energy from the incoming field and re-emits it. This is directly analogous to the effective aperture A_{eff} of a receiving antenna (§ 8):

$$A_{\text{eff}} = \frac{\lambda^2}{4\pi} G \quad \leftrightarrow \quad \sigma_T = \frac{8\pi}{3} r_e^2.$$

In both cases the effective area is not the geometric size of the object but the result of mode compatibility between incoming field and node

structure. For photons with $\hbar\omega \gg m_e c^2$, Compton scattering applies — the node structure of the electron can no longer be described as point-like and the cross-section falls with increasing energy.

FFGFT interpretation: node as local mode concentration

In FFGFT a particle is a local concentration of modes on the carrier $T^4/\mathbb{Z}_3$ — a location where specific winding numbers (n_θ, n_φ) are occupied and the field locally assumes a stable, non-trivial configuration **[POSIT]**. The near field of the particle is the immediate vicinity of this node in the carrier, projected onto the observation space; the far field is the radiation arising when the node changes its mode occupation.

The boundary between near and far field is therefore not a sharp line but the transition between the region where the node structure dominates the field and the region where the field propagates as a free wave. This transition lies at $r \sim \lambda_C$ for relativistic descriptions and at $r \sim a_0$ for atomic transitions — both are structural length scales of the node, not free parameters **[S]**.

9 Scale-invariance of the coupling condition: atoms, elements and planetary orbits

The principle from §8–10 — geometry of the carrier determines permitted coupling angles, coupling only under mode compatibility — is not restricted to antennas or individual particles. It holds on all physical scales. The scale changes; the structure of the condition remains identical. Doc. 242 (scale ladder) elaborated this connection at the cosmological level; here it is developed starting from antenna engineering.

Atoms: discrete modes in the Coulomb potential

The Schrödinger equation in the Coulomb potential has discrete eigenmodes, classified by quantum numbers (n, l, m) [13, 14]:

$$E_n = -\frac{m_e e^4}{2\hbar^2 n^2} = -\frac{13.6 \text{ eV}}{n^2}.$$

This is nothing other than the resonance condition of a carrier: the carrier is the Coulomb potential (spherically symmetric, $V \propto -1/r$);

the permitted modes are the solutions with integer n . The atom is a cavity resonator with spherical geometry — the electron node can only assume certain standing wave configurations, just as the TE_{mn} modes in a metal cavity (§ 9).

The selection rules $\Delta l = \pm 1$, $\Delta m = 0, \pm 1$ (§ 10) are the impedance-matching conditions of this resonator: a dipole photon ($l = 1$) couples only to transitions in which the change of node geometry is compatible with a dipole mode. A quadrupole photon ($l = 2$) couples with $\Delta l = \pm 2$ but more weakly by α^2 — because the mode overlap efficiency is smaller, identical to the aperture efficiency $\eta_A < 1$ of a horn antenna.

Periodic table: mode classification of a many-body system

The shell structure of the periodic table — $1s^2 2s^2 2p^6 3s^2 \dots$ — is a classification by mode geometries of the effective central potential [E]:

Shell n : $2n^2$ places (s, p, d, f with spin).

Each sub-shell (s, p, d, f) has a different orbital angular momentum l and thus a different spatial mode geometry. Chemical bonding arises through coupling of the boundary modes of two atoms: when the outermost occupied orbitals (HOMO/LUMO) have compatible geometry — sufficient spatial overlap, compatible symmetry — a covalent bond forms. The overlap integral

$$S_{AB} = \int \psi_A^*(\mathbf{r}) \psi_B(\mathbf{r}) d^3r$$

is the direct quantum-mechanical analogue of the polarisation factor $\cos^2 \psi$ in the Friis equation (§ 8): $S_{AB} = 0$ means no coupling, regardless of the energy of the atoms.

The noble gases (closed shells, $S_{AB} \approx 0$ for all bonding partners) are the quantum-mechanical analogue of orthogonal polarisation: no energy transfer because no mode compatibility.

Crystal lattice: the Bragg condition as compatibility condition

In a crystal the atoms form a periodic lattice of lattice constant a . An electromagnetic wave (X-ray, neutrons, electrons) is coherently reflected only when the Bragg condition is met [15, 16]:

$$2d \sin \theta = n\lambda, \quad n \in \mathbb{Z},$$

where d is the lattice plane spacing and θ the glancing angle. This is an integer phase condition: the path-length difference between partial waves reflected at successive lattice planes must be an integer multiple of the wavelength. Only then do amplitudes add constructively — identical to the phased-array condition (§ 8):

$$\phi_n = -nkd \cos \theta_0.$$

The crystal lattice is a natural phased array with fixed element spacing d . Bragg reflection selects exactly the angles of incidence at which mode compatibility holds; all other angles give destructive interference. In FFGFT the reciprocal lattice vectors $\mathbf{G}$ correspond to the permitted winding vectors of the carrier; Bragg reflection is the macroscopic manifestation of the quantisation condition on the torus **[S]**.

Molecular vibrations: infrared activity as a selection rule

A molecular vibrational mode is infrared-active (absorbs IR radiation) only when it produces a change in the electric dipole moment [17, 18]:

$$\left(\frac{\partial \vec{\mu}}{\partial Q} \right)_{Q=0} \neq 0,$$

where Q is the normal coordinate of the vibration. This is the impedance-matching condition between photon (dipole radiation) and vibrational mode: only when the vibration has a dipole geometry does it couple to a dipole photon. Symmetric stretching vibrations (as in N_2 or O_2) do not change the dipole moment — they are IR-inactive but Raman-active (coupling to quadrupole photons). This explains why N_2 and O_2 are not greenhouse gases: their vibrational modes are mode-incompatible with IR dipole radiation.

Planetary orbits: integer resonances in the gravitational field

The gravitational field of the sun is the carrier of the planetary system. Its modes are the Kepler orbits, classified by the orbital parameters (a, e, i) . When two planets are near an integer resonance $p : q$ ($p, q \in \mathbb{Z}$), small gravitational perturbations accumulate coherently over many revolutions (Doc. 242) [24, 19]. The Laplace resonance of the Galilean moons is the most famous example [19, 20]:

$$\frac{T_{Io}}{T_{Europa}} \approx \frac{1}{2}, \quad \frac{T_{Europa}}{T_{Ganymede}} \approx \frac{1}{2}, \quad n_{Io} - 3n_{Europa} + 2n_{Ganymede} = 0$$

(exact relation of the mean motions $n = 2\pi/T$). This relation is not a coincidence: it is a stable configuration because at exact resonance the mutual perturbations cause no secular energy input, only periodic. The system has an attractor at the integer resonance — identical to the attractor of an Arnold tongue in nonlinear dynamics (Doc. 328 § coupling regimes) **[K]**.

Further examples:

- Jupiter/Saturn: 5 : 2 resonance ("great inequality", known since Laplace 1785) **[Q]**.
- Neptune/Pluto: 3 : 2 resonance (Plutinos) **[Q]**.
- Kirkwood gaps in the asteroid belt: gaps at resonances 1 : 3, 1 : 4 with Jupiter (destructive resonance, not stabilising) **[Q]**.
- Saturn's rings: gaps (Cassini Division) at 2 : 1 resonance with Mimas **[Q]**.

The coupling condition is the same as in antenna engineering: integer phase compatibility. The carrier differs (gravitational rather than electromagnetic field); the structure of the condition is identical.

Summary table: coupling condition on all scales

Scale	Carrier	Coupling condition	Example	Analogue
Antenna	radiator	$\cos^2 \psi, Z$	Yagi	Friis
Atom	Coulomb	$\Delta l = \pm 1$	H lines	sel.rule
Molecule	potential	$\partial \mu / \partial Q \neq 0$	IR	dipole
Crystal	lattice	$2d \sin \theta = n\lambda$	Bragg	array
Per.table	shells	overlap S_{AB}	bond	aperture
Mol.bond	orbital	symm.com-pat.	HOMO	modes
Planets	gravity	$p n_1 = q n_2$	Laplace	Arnold

The common principle

All rows of the table describe the same thing: a system has a carrier with a mode spectrum; coupling to another system is only possible where the mode spectra are compatible; incompatible modes do not couple, regardless of the strength of the fields involved.

The technical maturity of antenna engineering — decades of measurements, standardised procedures, commercial simulators

— makes it the most tangible realisation of this principle. Atomic mechanics, solid-state physics and celestial mechanics are the same principle on different carriers. In FFGFT the common carrier is $T^4/\mathbb{Z}_3$; the coupling conditions on all scales are projections of winding-number compatibility onto the respective observation space **[S]**.

10 Duality: field and particle, energy and current

FFGFT treats time–mass duality as a fundamental principle (fundamental equation $\tilde{T} \cdot m = 1$, Doc. 365) **[POSIT]**. The same duality appears in electromagnetic field theory:

Field view		Current view
Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}$	$\longleftrightarrow$	$P = V \cdot I$
Field energy in dielectric	$\longleftrightarrow$	$I^2 R$ Joule loss
Evanescent wave outside	$\longleftrightarrow$	reactive power
Winding number (topological)	$\longleftrightarrow$	mass (dynamical)

In none of these pairs is one side “more fundamental” than the other — they describe the same field process from different projections. This is the precise analogue of time–mass duality in FFGFT: $\tilde{T}$ and m are reciprocal representations of the same object, not two different quantities **[K]**.

Reactive power and evanescent fields

On the G-line energy pulses back and forth in the periphery of the wire field — energy flux alternates direction in step with the field. This is reactive power in AC circuit theory, or — stated optically — the evanescent field on the shadow side of a totally reflecting interface **[E]**. In FFGFT this behaviour corresponds to the non-propagating part of the Fourier modes on $T^4/\mathbb{Z}_3$: modes below the spectral gap (Doc. 314 § H) carry no net energy but structure the spectrum **[S]**.

11 Exact agreement: Poynting integral and power formula

It is instructive to understand the exactness of this identity, because it is regularly presented in courses as an “approximate agreement” or a “plausibility argument”. Poynting’s theorem states **[E]**:

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{j} \cdot \mathbf{E},$$

where $u = \frac{1}{2}(\epsilon_0 E^2 + \mu_0^{-1} B^2)$ is the electromagnetic energy density. Integration over the volume of the cable with the corresponding boundary conditions yields exactly $P = V \cdot I$ —the only prerequisite is the validity of the Maxwell equations.

In FFGFT units (without ϵ_0, μ_0, e) the identity reduces to:

$$P = E \cdot m,$$

which follows immediately from $V = E$, $I = m$ and the fundamental equation $\tilde{T} \cdot m = 1$ **[K]**. The exactness is structural, not a numerical coincidence.

12 Summary

1. **Primary localisation.** Energy flows in the fields ($\mathbf{S} = \mathbf{E} \times \mathbf{H}$), not in the conductor. Voltage and current are macroscopic projections of the same field process **[E]**.
2. **Geometric objects.** Field lines on the carrier $T^4/\mathbb{Z}_3$ are topological objects with winding numbers; they classify modes and determine mass scales via $\tilde{T} \cdot m = 1$ **[B]**.
3. **FFGFT units.** With $c = \hbar = \alpha = 1$: $I = m$, $V = E$, $P = E \cdot m$ —exactly, not approximately. The Maxwell equations carry no prefactors **[K]**.
4. **G-line as illustration.** The Goubau experiment demonstrates macroscopically that energy transport without an explicit return conductor is possible because the transport occurs in the field. The cone transition corresponds to impedance matching; field lines locally optimise energy flux **[E]**.
5. **Duality.** Field view and current view are reciprocal projections — no pair is more fundamental. This mirrors the time–mass duality $\tilde{T} \cdot m = 1$ at the level of classical fields **[S]**.

References

Bibliography

- [1] G. Goubau, *Surface waves and their application to transmission lines*, J. Appl. Phys. **21** (1950) 1119–1128. Foundational work on single-wire wave guidance; cone transition, field configuration, loss measurement of 6 dB/mile.
- [2] F. Harms, *Elektromagnetische Wellen an einem Draht mit isolierender zylindrischer Hülle*, Ann. Phys. **23** (1907) 44–60. First analysis of the dielectric as a field-guiding layer; precursor of the Goubau line.
- [3] J. A. Stratton, *Electromagnetic Theory*, McGraw-Hill, New York, 1941. Standard reference for the Poynting theorem (Ch. 2), boundary conditions, field energy.
- [4] D. J. Griffiths, *Introduction to Electrodynamics*, 3rd ed., Prentice Hall, 1999. Poynting vector, Larmor formula, radiation resistance, reciprocity; accessible derivation of all standard results used in § 2–5.
- [5] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley, 1999. Larmor formula (Ch. 14), Thomson cross-section (Ch. 14.8), classical electron radius, Compton wavelength; canonical reference.
- [6] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed., Wiley, 2016. Standard text: radiation patterns, gain, radiation resistance, short dipole, half-wave dipole, monopole, Yagi-Uda, horn, parabolic, patch, phased array, Friis equation, reciprocity theorem.
- [7] H. T. Friis, *A note on a simple transmission formula*, Proc. IRE **34** (1946) 254–256. Original paper on the link budget equation;

"The received power is [...] proportional to the product of the transmitting and receiving gains."

- [8] H. Yagi, *Beam transmission of ultra short waves*, Proc. IRE **16** (1928) 715–741. Original Yagi–Uda paper; reflector and director principle; gain through phase interference of passive elements.
- [9] D. M. Pozar, *Microwave Engineering*, 4th ed., Wiley, 2012. Cavity resonator (Ch. 6), patch antenna (Ch. 14), horn antenna, waveguide modes, impedance matching, Smith chart, cavity perturbation.
- [10] R. J. Mailloux, *Phased Array Antenna Handbook*, 3rd ed., Artech House, 2017. Array factor, beam steering, mutual coupling, active impedance matrix, grating-lobe condition; quantitative treatment of all array phenomena.
- [11] S. A. Maier, *Plasmonics: Fundamentals and Applications*, Springer, 2007. Surface plasmon polaritons, dispersion relation (Ch. 2), coupling condition, near-field enhancement; standard text of nanophotonics.
- [12] A. Kurs, A. Karalis, R. Moffatt, J. D. Joannopoulos, P. Fisher, M. Soljačić, *Wireless power transfer via strongly coupled magnetic resonances*, Science **317** (2007) 83–86. Resonant near-field coupling over 2 m with 40 % efficiency; *"The transfer is mediated by the overlap of the near fields."* Direct evidence for energy transfer without radiation through mode compatibility.
- [13] D. J. Griffiths, *Introduction to Quantum Mechanics*, Prentice Hall, 1995. Hydrogen atom, quantum numbers, selection rules $\Delta l = \pm 1$, $\Delta m = 0, \pm 1$ (Ch. 9); overlap integral and bonding theory (Ch. 7).
- [14] C. Cohen-Tannoudji, B. Diu, F. Laloë, *Quantum Mechanics*, 2 vols., Wiley, 1977. Selection rules (Ch. XIII), dipole approximation, quadrupole suppression $\sim \alpha^2$; canonical reference for atomic transitions.
- [15] C. Kittel, *Introduction to Solid State Physics*, 8th ed., Wiley, 2005. Bragg condition (Ch. 2), reciprocal lattice, Brillouin zone, lattice dynamics; standard solid-state text.

- [16] N. W. Ashcroft, N. D. Mermin, *Solid State Physics*, Holt, Rinehart and Winston, 1976. Bragg scattering, phased-array analogy of reciprocal lattice (Ch. 6), electron bands; advanced reference.
- [17] P. Atkins, J. de Paula, *Physical Chemistry*, 9th ed., Oxford University Press, 2010. IR activity condition $\partial \vec{\mu} / \partial Q \neq 0$ (Ch. 13), Raman activity, symmetry selection rules; standard physical chemistry text.
- [18] E. B. Wilson, J. C. Decius, P. C. Cross, *Molecular Vibrations*, McGraw-Hill, 1955. Normal coordinates, selection rules, group theory of vibrational modes; classic original reference.
- [19] C. D. Murray, S. F. Dermott, *Solar System Dynamics*, Cambridge University Press, 1999. Orbital resonances (Ch. 8), Laplace resonance Io/Europa/Ganymede, Kirkwood gaps, Cassini Division, Arnold tongues in celestial mechanics; authoritative modern reference.
- [20] S. J. Peale, P. Cassen, R. T. Reynolds, *Melting of Io by tidal dissipation*, *Science* **203** (1979) 892–894. Laplace resonance as the cause of tidal heating of Io; “*The forced eccentricity [...] results from the Laplace relation $n_1 - 3n_2 + 2n_3 = 0$.*” Spectacular prediction before Voyager 1 arrival.
- [21] J. Pascher, *Doc. 365: The Foundation of FFGFT in Four Layers — fundamental equation, electromagnetic sector, scaling parameter and algebraic confirmations*, FFGFT corpus, Zenodo (2026). Layer 1 ($\vec{T} \cdot m = 1$), Layer 2 (FFGFT units, $I = m$, $V = E$, Maxwell without prefactors), Layer 3 ($\xi = 4/30000$).
- [22] J. Pascher, *Doc. 314: Lattice in Hilbert Space*, FFGFT corpus, Zenodo (2026). Fourier modes on $T^4/\mathbb{Z}_3$, winding numbers as topological invariants, shell count to norm 120, spectral gap.
- [23] J. Pascher, *Doc. 328: Coupling regimes, resonance and synchronisation*, FFGFT corpus, Zenodo (2026). Galois lattice condition, Arnold tongues, Adler model, $K_{\text{eff}} = 2\pi\xi$, coupling threshold; direct connection to orbital resonances and mutual coupling.

- [24] J. Pascher, *Doc. 242: Scale ladder — interscale couplings in FFGFT*, FFGFT corpus, Zenodo (2026). Planetary orbit argument, Laplace resonance as structural principle, scale separation as historical convention, not physical necessity.
- [25] J. Pascher, *Doc. 124: Charge as geometric projection*, FFGFT corpus, Zenodo (2026). Charge as emergent projection of geometric patterns; particles as field configurations, not intrinsic point properties.
- [26] J. Pascher, *Doc. 343: The zeta function of the $T^4/\mathbb{Z}_3$ torus*, FFGFT corpus, Zenodo (2026). Resolution floor 100ξ , Galois lattice, Farey critical denominator height; quantitative bound on mode distinguishability.